# Rotated evaluation of the 31+32τ radix step

**Question.** Can the sparse 31+32τ Horner step use the secp256k1 order-three x rotation to replace explicit τ evaluation, and does that schedule change measured cold costs?

**Scope and instance.** This is an implementation comparison on SEC 2 secp256k1 ($E: y^2 = x^3 + 7$ over $p = 2^{256} - 2^{32} - 977$, prime subgroup order n=FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD03 The frozen input fixture has 12,288 scalar/point pairs. Its SHA-256 is:

`a1c4964f174e36ea26f6a1897128f651a2ce7251e4bf4ffd532ff09221e721d3`

No new curve or scalar rule is proposed.

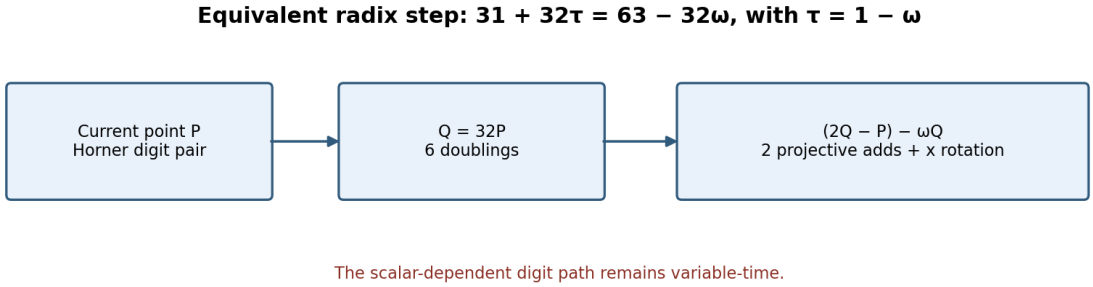



Figure 1: Rotated radix step method

**Identity and native review.** In the pair basis, τ=1−ω, so 31+32τ=63−32ω. With Q=32P, the native candidate evaluates (2Q−P)−ωQ per nonidentity Horner step: six doublings, two projective additions, and one x rotation. The hash-guarded overlay changes only this radix step and the pilot entry point relative to the adaptive digit graph.

The pinned Lean module proves τ=1−ω, the pair-lattice identity 63−32ω=31+32τ, and its transfer under explicit addition, subtraction, scaling, and endomorphism-map hypotheses. I reran the Lean checker with the pinned 4.34.1 release; the dated fresh receipt preserves source hashes and `#print axioms` for all three declarations. The module does not prove the native x-rotation implementation or curve-coordinate/group-law refinement; the fixture replay checks that implementation on a finite workload.

The radix digit selection remains scalar-dependent, as in the preceding graph implementation. The path is **not constant-time** and must not be used for secret-scalar multiplication.

**Matched Tier 1 panels.** The one-scalar and 128-point plans compare the rotated step with the digit-graph path on identical frozen inputs, RunPod pod `thyvmdqmzsz5ik`, CPU 40, 16 alternating repetitions, and adjacent reference/reference A/A controls. Each panel completed 96 arms and 48 pairs. Rust `Instant` includes scalar-local preparation and Horner evaluation; fixture loading, process launch, and external replay are outside the timed interval. The one-scalar rules, launch, and raw archive identify job `c61069334c8a4d83bebbb05dc242e1b2`. The 128-point rules, launch, and raw archive identify job `93837f49c72c4f0d9311366066628dc7`. The native provenance retains both exact source trees and binaries.

| Frozen workload | Graph reference mean | Rotated candidate mean | Qualified paired median | Paired bootstrap 95% interval |
|---|---|---|---|---|
| One scalar | 0.561 ms | 0.544 ms | 1.012× | [1.003, 1.035] |
| 128 points | 39.898 ms | 39.170 ms | 1.021× | [1.010, 1.025] |

The independent auditor checked hashes, raw commands and output, pair records, independent scalar replay of every measured fixture point, artifact stability, and all A/A controls. Both panels were qualified

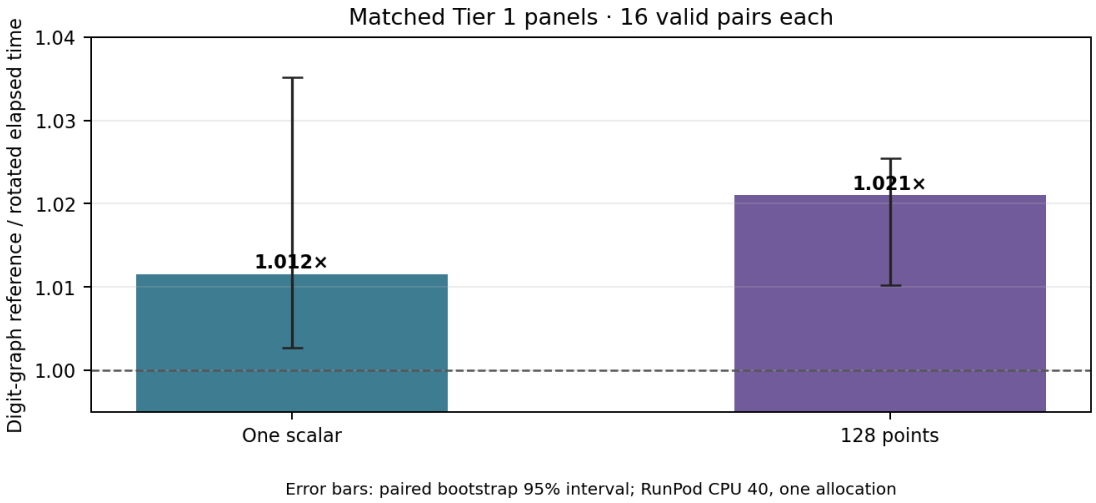



Figure 2: Matched rotated-step ratios

with no invalid arms or A/A failures. These are small Tier 1 observations on one allocation, not a general performance guarantee.

**Full-fixture replay.** The frozen manifest, launch, and raw archive preserve this run:

`09eeb419886a43e393492b4511a8c233`

Both identical candidate arms verified all 12,288 outputs; the auditor checked the fixture, binary, raw commands, pair ledger, and independent final-scalar replay. Arm times were 3.699 s and 3.712 s. This A/A replay has a null speedup claim and proves neither all-scalar correctness nor a matched full-stream speedup.

**Graph status and remaining bridge.** The step is an implementation transform for an existing curve and changes no canonical curve/isogeny node or route. I checked the experiment's adaptive digit graph study and the repository curve catalog/route records; no graphable fact changed. The shared sparse-radix point-correctness problem remains open for the selector, native $\omega/\tau$ maps, and curve group-law refinement.

The editable figure source, SVG and 180-dpi PNG copies, and this report's PDF are committed together.